

Virtual Teammate — Backend Side Documentation

The complete technical reference: every plugin, the custom code (plugins, mu-plugins, WPCode snippets), the data model, roles, the HubSpot sync engine, the member portal, and integrations. Companion doc: [DOCS-MARKETING.md](#) (public site, funnels, content). **Site:** the **legacy (old) virtualteammate.com** WordPress website — <https://virtualteammate.com> — the prior WordPress build being superseded by the new `vtnew` marketing site. DB `0k7.183.mytemp.website_1772483989`, table prefix `wpvividstg01_`.

0. Stack at a glance

Layer	Technology
CMS	WordPress (schema <code>db_version</code> 61833, ~WP 6.7-era)
Theme	<code>hello-elementor</code> 3.4.7 (stock, no child theme)
Page builder	Elementor 4.1.1 + Elementor Pro 4.0.1
Membership / roles / login	Ultimate Member 2.11.4
Data fields	ACF 6.8.2 + CPT-UI 1.19.2
CRM / marketing	HubSpot via LeadIn 11.3.45 + custom <code>vt-hubspot-user-sync</code> 1.1.0
Forms	WPForms 1.10.1
Video	Presto Player 4.2.0
Security / backup / cache	Wordfence 8.2.2, WPVivid 0.9.128, LiteSpeed Cache 7.8.1
Custom code	2 custom plugins, 15 mu-plugins, 26 active WPCode snippets

Users: 169 total — `um_vt-hired` 76, `um_vt-onpool` 43, `um_clients` 37, `administrator` 4, `um_csm` 3, `editor` 2, `subscriber` 2, `um_ambassador` 1.

⚠ **Architectural note:** almost all custom behavior lives in **mu-plugins** and **WPCode snippets** (auto-inserted PHP), **not** in a child theme. Some snippet code is loaded via `eval()` from the database by the mu-plugins. The theme is stock and disposable.

1. Plugins

Active (13)

Plugin (slug)	Ver	What it does here
Advanced Custom Fields (<code>advanced-custom-fields</code>)	6.8.2	The structured data layer behind VA profiles (rates, badges, photos, HubSpot IDs). 4 field groups (\$5).
Custom Post Type UI (<code>custom-post-type-ui</code>)	1.19.2	Registers the public profile CPT <code>vt-list-by-category</code> . (Other CPTs are registered in code.)
Elementor (<code>elementor</code>)	4.1.1	Visual page builder for all marketing pages.
Elementor Pro (<code>elementor-pro</code>)	4.0.1	Theme Builder (headers/footers/dynamic profile templates), Pro widgets, popups.
If Menu (<code>if-menu</code>)	0.19.2	Conditional menu visibility by login state / role.
LeadIn (HubSpot) (<code>leadin</code>)	11.3.45	Official HubSpot: tracking script, form capture, CRM sync (OAuth).
LiteSpeed Cache (<code>litespeed-cache</code>)	7.8.1	Page caching + CSS/JS/image optimization.
Presto Player (<code>presto-player</code>)	4.2.0	Video player (VA intros, courses), incl. BunnyCDN/HLS.
Ultimate Member (<code>ultimate-member</code>)	2.11.4	Front-end login/registration/profiles + the role system (the portal).
Virtual Teammate Plugin (<code>vt-hubspot-user-sync</code>)	1.1.0	The custom core. HubSpot↔WP sync, client/VT/CSM provisioning, dashboard chat, notifications. \$3.
Wordfence (<code>wordfence</code>)	8.2.2	Firewall, login protection, malware scan.
WPForms (<code>wpforms</code>)	1.10.1	Lead-capture forms.
WPVivid Backup (<code>wpvivid-backuprestore</code>)	0.9.128	Scheduled backups + site cloning/migration.

Installed but inactive (6)

Plugin (slug)	Ver	Note
Better Search Replace (<code>better-search-replace</code>)	1.4.10	DB URL find/replace — run on demand to normalize any hardcoded dev URLs to <code>https://virtualteammate.com</code> .
WPCode Lite (<code>insert-headers-and-footers</code>)	2.3.6	⚠ The WPCode snippets still exist and run even though this manager plugin is inactive — the mu-plugins load snippet code directly. Don't assume "inactive" = snippets off.
SQLite Object Cache (<code>sqlite-object-cache</code>)	1.6.3	Optional persistent object cache; off here.
VT HTML Site Importer (<code>vt-html-site-importer</code>)	1.0.0	Custom one-off migration tool (\$4).
Wise Chat with AI (<code>wise-chat</code>)	3.4	Inactive as a normal plugin, but its engine runs as a mu-plugin (<code>0-wise-chat-engine.php</code>).
Asset CleanUp (<code>wp-asset-clean-up</code>)	1.4.0.4	Per-page asset trimming; off (LiteSpeed handles this).

2. Custom mu-plugins (always-active)

Location: `wp-content/mu-plugins/` . Must-use plugins load on every request and **cannot be deactivated from admin**. They load alphabetically (`0-` first, `zzz-` last). Custom code uses the `codex_` function prefix.

2.1 `0-wise-chat-engine.php` (*vendor*)

High-performance standalone AJAX endpoint for Wise Chat; inert unless `?wc-gold-engine` is set. ⚠ Contains a **hardcoded absolute Windows filesystem path** to the Wise Chat `Loader.php` that breaks on a Linux host — must be repointed per environment.

2.2 `automation-by-installatron.php` (*vendor*)

Disables WP auto-updates + the related Site Health test (updates managed by the host's Installatron).

2.3 `client-dashboard-access.php` — **portal login/redirect router**

The central access controller. Detects login/dashboard requests, disables caching on them, gates which UM login form each role may use, and redirects logged-in users to their role dashboard.

- **Role → dashboard map:** clients → `/user-2/`; VTs (`um_vt-hired` / `um_vt-onpool`) → `/vt-profile/`; CSMs (`um_csm`) → `/csm-profile/` .
- **Login form-ID → allowed-role map** (hardcoded): `39` →subscriber/clients; `1131` →ambassador/vt-hired/vt-onpool; `1993` →vt-hired/vt-onpool; `1927` , `1636` →applicant; `2680` →wpseo_manager/csm.
- Swaps the logged-in header menu to "My Account / Log Out", rewrites legacy applicant-login menu items, renames post 1137 → "VT Hired Login".
- Disables UM login nonce verification; injects an autoplay-blocking script on login pages.

2.4 `client-resources-shortcode.php`

Render function for `[client_resources]` (a Downloads grid of client PDFs/worksheets). Tag registered by `dashboard-snippet-fallbacks.php` . Hardcoded Wix CDN PDF URLs + local cover images + CTA `/client-playbook-3/` .

2.5 `dashboard-roi-chat-fixes.php`

Two jobs: (1) on every front-end load, ensures `va_conversation` chat threads exist between a client and their linked VAs/CSMs/reviewer; (2) overrides the ROI calculator's `roix_get_contracts` AJAX (priority 1) to resolve the client's company from HubSpot user-meta and pull contracts from the HubSpot CRM. Requires `HUBSPOT_PRIVATE_APP_TOKEN` ; hardcoded HubSpot custom object `2-31153232` .

2.6 `dashboard-snippet-fallbacks.php` — the portal orchestrator (~12,500 lines)

The heart of the client/VT/CSM portal. On `plugins_loaded` it:

- `require_once` s the four sibling shortcode files (`roi-savings-calculator` , `resource-hub` , `vtm-hr-hub` , `client-resources`) — **this is the only place those shortcodes get registered.**
- Re-executes inactive WPCode snippets by ID via `eval()` if their shortcode is missing (hardcoded IDs: `6106` , `6402` , `5956` , `3278` , `6026` , `1551` , `5957` , `3028` , `3269` , `6020` , `6021` , `3099` , `3500` , `2196`).
- Registers ~25 portal shortcodes (each `remove_shortcode` first so PHP wins): `[vt_taskmanagement]` , `[vt_assignments]` , `[select_va_button]` , `[special_link_generator]` , `[vt_my_csm_vt]` , `[manager_va_dashboard]` , `[roi_savings_calculator]` , `[selected_va_list]` , `[vt_talent_pool]` , `[vt_request_pool]` , `[csm_workday_tracker]` , `[hubspot_invoice_viewer]` , `[workday_tracker_profiles]` , `[vtm_resource_hub_consolidated]` / `[vtm_help_center]` , `[vtm_hr_hub]` , `[client_resources]` , `[my_payslip]` , `[va_workday_tracker]` , `[va_booking_center]` , and more.
- Installs a custom table `{prefix}codex_eod_reports` (end-of-day reports), schedules a **daily EOD reminder cron** (`codex_eod_daily_reminder_event`), registers a booking CPT, and wires AJAX for VA selection, CSM approvals, EOD save, and CSM special links.
- **CSM special links:** token-based temporary talent-pool access (option `codex_csm_special_links`); invalid/expired tokens `wp_die` 410.
- Defines 210+ `codex_*` functions across EOD reports, talent pool, VA selection, booking, task management, payslips.

2.7 `elementor-safe-mode.php` (*vendor*)

Standard Elementor Safe Mode (editor-only isolation troubleshooting).

2.8 `resource-hub-shortcode.php`

Render function for the **Help Center** hub (`[vtm_resource_hub_consolidated]` / `[vtm_help_center]`) — searchable card grid linking to Knowledge Center, HR, Finance, IT, Marketing, Community. Community tile hardcoded to `https://virtualteammate.com/group-teammate-community-about/`.

2.9 `roi-savings-calculator-shortcode.php`

Backs `[roi_savings_calculator]` (the "Value Creation Calculator"). Resolves the client's company via HubSpot contact, then Google-Sheet fallbacks, then pulls VT contracts/deals from HubSpot. **Default original** `roix_get_contracts` **AJAX handler (priority 10)**.

- ⚠️ **Hardcoded Google Sheet** `1sRpFUHuf5QR3av0nzRErDXGREwzdcJSwu7SfGU9_Nqk` (tabs `632565396` Clients, `1424706697` VT Accounts).
- ⚠️ A **commented-out live-looking HubSpot token** (`//$token = "pat-na1-77aafd22-..."`) is left in source — should be scrubbed.

2.10 `um-role-sync.php` — **role guarantor**

On every `init`, ensures the four UM custom roles exist as both WP roles and UM roles:

- `um_vt-hired` ("VT Hired"), `um_vt-onpool` ("VT Onpool"), `um_clients` ("Clients"), `um_csm` ("CSM").
- `um_csm` is granted the `wpseo_manager` capability and is treated as a "manager." Also retroactively promotes existing `wpseo_manager` users to `um_csm` (batched 200).

2.11 `vtm-dashboard-redirect.php`

Shortcode `[vtm_dashboard_redirect]` — put on a login page to bounce logged-in users to their role dashboard (server-side redirect + JS fallback). Defaults: client `/user-2/`, vt `/vt-profile/`, csm `/csm-profile/`.

2.12 `vtm-demo-account.php` — **demo data provisioner**

Class `VTM_Demo_Account` seeds a self-contained demo client (`demo@virtualteammate.com`) with fictional VTs, conversations, EOD reports, and mock ROI data, hidden from public listings. Admin-only AJAX (seed/reset/topup, cap `manage_options`). Overrides `roix_get_contracts` at **priority 0** (mock contracts for the demo client).

- ⚠️ **Plaintext demo credentials committed in source** — all accounts use password `demo12345` (`demo@`, `csm.demo@`, `onpool.demo@`, `maya.demo@`, `carlos.demo@`, `aisha.demo@virtualteammate.com`). Flagged for production.

2.13 `vtonpool-hide-tabs.php`

Shortcode `[vtonpool_hidetabs]` — for on-pool-only VTs, hides dashboard tabs (Payslips, Meetings, Workday Tracker, My Assigned Tasks, My CSM, Virtual Teammate) via CSS/JS + MutationObserver.

2.14 `vtm-hr-hub-shortcode.php`

Render function for `[vtm_hr_hub]` (HR Hub UI). Tag registered by the orchestrator.

2.15 `zzz-local-performance.php` — **development-only**

Entire file is a no-op unless the request comes from a local development host — it never runs on the live site. In a dev environment it seeds the PHP-version check transient, dequeues head clutter + unused UM/Presto assets, **blocks slow remote calls** (Elementor mixpanel/license, WordPress serve-happy), and rewrites Google Docs PDF viewers to direct URLs. No production effect.

Cross-file gotchas

- **Three stacked `roix_get_contracts` AJAX handlers** by priority: demo (0) → HubSpot-meta override (1) → original Sheet-based (10). First to call `wp_send_json_*` wins.
- Shortcode registration is **centralized** in `dashboard-snippet-fallbacks.php`; if it's renamed/disabled, the client-resources/help-center/HR-hub/ROI shortcodes vanish.
- Multiple files `eval()` WPCode post content from the DB — a maintenance + security consideration.

3. The custom core: `vt-hubspot-user-sync` (v1.1.0)

Main file ~11,300 lines; class `VT_HubSpot_User_Sync` (static) + `VT_Dashboard_Chat` + `VT_Notification_Center`. Despite the name it does **four** jobs:

3.1 Talent sync (HubSpot contacts → public VA profiles)

`run_sync()` (also runs **hourly via cron** `vt_hubspot_user_sync_cron`):

1. Lock (`vt_hubspot_user_sync_lock`, 45 min) → resolve token → bail if missing.
2. `search_virtual_teammate_contacts()` — HubSpot contacts where `hs_lead_status = "Virtual Teammate"`, paginated.
3. `hydrate_contacts()` — batch-read extra properties.
4. Per contact → `sync_contact_to_user()`: map role from `vt_status` (`hired` / `contracted` → `um_vt-hired`; `unmatched/eligible` / `matched` → `um_vt-onpool`; `no longer eligible` → delete), upsert WP user, write meta, import profile photo.
5. `upsert_public_profile_posts()` → creates/updates `vt-list` (card) + `vt-list-by-category` (full profile) posts, imports media into a `vtmedia` uploads area.

6. Cleanup stale/duplicate posts + orphan media → purge caches.

3.2 Client sync (HubSpot companies → portal clients) — batched state machine

Stages (run incrementally via AJAX `ajax_client_sync_step`, state in options): `init` → `hired_contacts` → `csm_owners` → `companies` → `inactive` → `cleanup_vtmedia` → `finalize`.

- Reads companies with lead status "**Client - Active**", walks company→contact, company→CSM, and a custom **contract object** `2-31153232` (association type `28`).
- `sync_company_to_client_user()` provisions the client WP user (role `um_clients`) and links their VTs + CSMs.
- `deactivate_inactive_client_user()` handles "Client - Inactive" — detaches conversations, clears notifications, drops the contract cache row.
- Contract data cached in custom table `vtcontracts`.

There are parallel **batched talent sync** and **Purge VT Data** state machines (same stage pattern).

3.3 Dashboard tooling

- **VT_Dashboard_Chat** — front-end polling chat (client / assistant / manager) on `va_conversation` + `va_message`. Shortcodes `[va_chat_window]`, `[va_assistant_chat]`, `[va_manager_chat]`. ⚠️ Bootstraps by `eval()`-ing **WPCode snippets 2104 & 2105** if helper fns aren't defined.
- **VT_Notification_Center** — in-app + email notifications stored in user-meta `va_notifications`; shortcode `[va_notification_center]`. Email via native `wp_mail()`.
- Client meeting scheduling (`va_meeting`), "request additional VT" cards, assignment-roster sync (`vtm_assignment`).

3.4 Admin UI

Users → VT HubSpot Sync (cap `manage_options`). Tabs: Talents, Clients, Reset, **Purge VT Data** (destructive, batched), Sync Report, Demo Data.

3.5 Integrations & credentials

- **HubSpot CRM v3 REST** (`https://api.hubapi.com`, Bearer). Token resolution order: DB option `vt_hubspot_user_sync_settings['token']` → constant `HUBSPOT_PRIVATE_APP_TOKEN` → constant `HUBSPOT_PRIVATE_TOKEN`.
 - **Google Drive** profile-photo fallback (public folder `1MBKL1oD0S01Ro-uIYzYAyE3un1qn7CJD`, no creds).
 - **Email**: native `wp_mail()` only (no API).
 - Hardcoded portal-specific IDs: contract object `2-31153232`, association type `28`, table `vtcontracts`, chat snippet IDs `2104 / 2105`.
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4. vt-html-site-importer (v1.0.0, inactive)

One-off migration tool. **Tools** → **HTML Site Importer** (cap `manage_options`). Imports a static/PHP site (folder of `.html` or a `sitemap.xml`) into WP Pages: extracts `<main>` / `<article>` content, copies assets into `uploads/vt-imported-site/<slug>/` with URL rewriting, idempotent re-runs (matched by meta `_vt_html_import_source`), supports dry-run, status, slug prefix, host rewrite, max 200 pages. No REST/AJAX/cron; runs synchronously via `admin-post.php`. No hardcoded env values.

5. Data model

Custom Post Types

CPT-UI registers only one: `vt-list-by-category` ("VT View Profiles") — the public VA profile directory. **All others are registered in code** (WPCode snippets / plugin):

Post type	Count	Purpose
<code>pp_video_block</code>	524	Presto Player video blocks
<code>va_conversation</code>	120	Chat threads
<code>vt-list</code>	117	VA listing cards (internal companion)
<code>vt-list-by-category</code>	117	Public VA profiles
<code>va_message</code>	42	Chat messages
<code>vtm_assignment</code>	16	Client↔VA assignments
<code>um_form</code>	13	Ultimate Member forms
<code>va_meeting</code>	3	Scheduled meetings
<code>um_directory</code> , <code>va-profile</code> , <code>vt_showcase</code>	1 each	Directory / legacy profile / showcase
<code>virtual_assistant</code>	(code)	Registered in WPCode snippet 2020

ACF field groups (4)

- **VT Profile** (key `vt-profile`) — public profile fields: `department`, `country`, `skills`, `primary_roles`, 6 badge images (HIPAA, EF, DISC, predictive, AT, IQ) + captions.
- **View Profiles** (key `view-profiles`) — intake/editable: `profile_picture`, `name`, `department`, `summary`, `experience`, 5 badge slots, `upload_video`, `upload_resume`, `email`, `link_users` (links to WP user).
- **Account Sync** (key `account-sync`) — `enter_email`, `linked_user`, `va_manager_id`.

- **Business Selection for Visibility** (key `business-selection-for-visibility-purpose`) — `medical_niche` (radio); drives medical/non-medical content gating (snippet 5846).

Custom DB tables

- `{prefix}codex_eod_reports` — end-of-day reports (created by the orchestrator mu-plugin).
- `{prefix}vtcontracts` — cached HubSpot contract data (created by the sync plugin on activation).

6. Roles & access

Roles (WP `user_roles` option)

Standard: administrator, editor, author, contributor, subscriber. Custom: `ambassador`, `applicant`, `prospect_viewer`, `vtm-admin` (VTM Admin), `vt-hired` (VT Hired), `vt-onpool` (VT Onpool), `clients` (Clients), `csm` (CSM).

UM prefixes its role keys `um_` when applied to users (e.g. `um_vt-hired`). `um_csm` carries the `wpseo_manager` capability and is treated as a manager by `va_is_manager()`.

Ultimate Member forms (login/register/profile)

Multiple login forms exist because each is gated to a role (WPCode snippet 3228 + the mu-plugin form→role map):

ID	Form	Mode	Registers role
38/39/40	Default Register/Login/Profile	—	—
1125/1131	VT Profile / VT Login	profile/login	—
1128	VT Registration	register	<code>um_ambassador</code>
1632/1636	Applicant Register / Log In	register/login	—
2680/2792/2794	Manager Login / Profile / Register	—	<code>wpseo_manager</code>
6285	Admin login	login	—

Portal dashboards by role

- **Client** → `/user-2/` (login `/client-login/`)
- **VT (hired/onpool)** → `/vt-profile/` (login `/vt-login/`)
- **CSM** → `/csm-profile/` (login `/csm-log-in/`)
- **Manager/Admin** → manager dashboard / admin pages

7. WPCode snippets (custom code in the DB)

Stored as `post_type='wpcode'` , auto-inserted PHP. **26 published (active), 38 drafts (inactive — many duplicates/experiments)**. ⚠ The WPCode manager plugin is "inactive" but snippets still run (mu-plugins load them, some via `eval()`). Key active snippets:

ID	Title	Role
5846	Business Niche Visibility	<code>body_class</code> filter for medical/non-medical content gating
2086	For Video Upload	<code>[acf_presto_video]</code> — BunnyCDN video
6192	Help Center (Courses)	<code>[vtm_help_center]</code>
6183	HR Content Help Center	<code>[vtm_hr_hub]</code>
2109	LightBox style video	<code>[vt_acf_video]</code>
2020	Loop Grid Filter	registers <code>virtual_assistant</code> CPT
6402	New ROI Calculator	<code>[roi_savings_calculator]</code> (the live one)
2849	Payslip and Report	<code>[admin_upload_payslip]</code>
6020	Refer a Friend	<code>[refer_a_friend]</code> + AJAX email
6026	Referral UI Client	<code>[vtm_referrals_tab]</code>
5851	Request To Join Modal	CTA modal + AJAX email
5957	Resources Tab	<code>[vtm_resource_hub_consolidated]</code>
2071	Resume Upload	embeds ACF resume PDF
5666	retrictclient	restricts <code>vt-page</code> , <code>careers</code>
3228	Role-based Login Restriction	maps UM login form → allowed role
2104	Select The VA	VA-assignment core (CPTs + AJAX)
3797	sharecommunityshortcode	<code>[share_group_popup]</code> ⚠ contains a hardcoded dev URL — repoint to <code>virtualteammate.com</code>
5692	va dashboard tracker	<code>[va_workday_tracker]</code>
6579	VT Prefilled Profile	UM default avatar/cover from ACF
6284	VT Resume & Video Restriction	gates Presto video on VA pages
3200	Workday Request Shortcode	client workday-report requests + AJAX
3278	workdaytracker shortcode	<code>[workday_tracker_profiles]</code> (CSM-approval)

8. Scheduled tasks (cron)

Event	Schedule	Source	Action
<code>vt_hubspot_user_sync_cron</code>	hourly	sync plugin	Talent sync from HubSpot
<code>codex_eod_daily_reminder_event</code>	daily	orchestrator mu-plugin	EOD report reminder

Client/talent/purge syncs are **AJAX-batched** (admin-triggered), not cron.

9. Integration & credential map

Integration	Where config / credentials live
HubSpot CRM (custom sync)	DB option <code>vt_hubspot_user_sync_settings['token']</code> or wp-config constant <code>HUBSPOT_PRIVATE_APP_TOKEN</code> / <code>HUBSPOT_PRIVATE_TOKEN</code> . Portal-specific: custom object <code>2-31153232</code> , assoc type <code>28</code> .
HubSpot (LeadIn/tracking)	OAuth — options <code>leadin_portalId</code> , <code>leadin_refresh_token</code> (secret).
Google Sheets (ROI calc)	Hardcoded sheet <code>1sRpFUHuf5QR3avOnzRErDXGREwzdcJSwu7SfGU9_Nqk</code> .
Google Drive (photos)	Public folder <code>1MBKL1oD0S01Ro-uIYzYAyE3un1qn7CJD</code> (no creds).
Email	Native <code>wp_mail()</code> (server mail / SMTP). See §11.
BunnyCDN (video)	Referenced in ACF <code>upload_video</code> per profile.

10. ⚠ Known issues & handoff checklist

1. **URL cleanup:** use **Better Search Replace** to normalize any remaining hardcoded dev URLs → `https://virtualteammate.com` across the DB. (One bad mangled-URL variant has already been cleaned from posts + Elementor data.)
2. **Fix mangled values:** WPForms form 6524's broken recipient address; snippet 3797's hardcoded share link.
3. **Hardcoded Windows path:** `0-wise-chat-engine.php` references an absolute `C:\...` filesystem path that will break on a Linux host.

4. **Secrets in source:** scrub the commented HubSpot token in `roi-savings-calculator-shortcode.php`; rotate if it was ever real. Demo creds (`demo12345`) are plaintext in `vtm-demo-account.php`.
 5. **No child theme:** the custom UM welcome email override (`hello-elementor/ultimate-member/email/welcome_email.php` , contains plaintext group-chat password `ClientVOnly`) will be **wiped on theme update**. Move to a child theme or plugin.
 6. **eval() of DB snippets:** several mu-plugins execute WPCode post content via `eval()` — audit before trusting on production.
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11. Email / SMTP

Portal notifications and UM emails send via WordPress `wp_mail()` from `support@ / nricamora@virtualteammate.com`. The domain mailbox is **Google Workspace**, which requires an **App Password** for SMTP (regular password won't authenticate). See [the email setup note](#) — the active app password and SMTP host/port belong in the SMTP plugin or wp-config, never committed to a tracked file.